

Improving Radar-Based Precipitation Nowcasts with Machine Learning Using an Approach Based on Random Forest

Andy Liu, October 7, 2026

(Mostly introduction and logistics)

Logistics

- Pizza Courtesy of Alex Turner
- MLJC Maintainers: Eric, Will, Andy
- Biweekly Meeting, Thursday 12-1pm @ ATG 311
- Fall Quarter Schedule
 - October 8
 - October 22
 - November 5
 - November 19
 - December 3

Logistics

- MLJC Website:
 - <https://nightingale-lzh.github.io/UW-Atmos-MLJC/>
 - Collection of previous paper discussions, workshops, etc.
- (I need someone to help me maintain the website. Ideally, first-year students)



Icebreaker & Introduction

- Name, Department, Year, Research Topic
- Summer Highlights, Hobbies(?)
- Survey:
 - What aspects of AI/ML do you like the most?
 - Math behind ML (linalg, stats, theory) x5, satellite x2, not training, XAI, linear systems, DMD, operational forecast, observations, DA, not math, ML on improving science, emulator, capability of ML model (speed, backprop), generative model
 - What AI/ML topics do you want to know about?
 - Generative model x2, not architecture, climate and AI x2, ML methods x2, benchmarking (evaluation) x3, process/dynamics emulation, ROM, uncertainty quantification, sampling, PDE problems

The paper we have today is not very important. We can take long time to introduce yourselves here

Fall Quarter Logistics

- October 8
 - Lead: Andy, Pizza: Eric
- October 22
 - Lead: Anna, Pizza: Ben
- November 5
 - Lead: August, Pizza: Will
- November 19
 - Lead: Chris, Yi, Sydney, Pizza: Hikari
- December 3 (week before AGU)
 - Lead: Will, Pizza: Chris
- Paper Discussion
 - We have a backlog of good papers.
 - Let Andy know if you have any paper suggestions
- Workshop
 - New workshop or repeat previous one
- Beta-test presentation
 - Ideally ML-related
- The lead has the final decision

Paper Discussion

- Improving Radar-Based Precipitation Nowcasts with Machine Learning Using an Approach Based on Random Forest
- Mao & Sorteberg, 2020
- Link: <https://doi.org/10.1175/WAF-D-20-0080.1>



Overview

- Norway
- Classification Problem
 - No precip vs. precip
 - Heavily skewed in one class (7% precip cases)
- Method: Random Forest
- Goal:
 - Improve radar and NWP
 - Generalizability across regions
 - Find good predictors

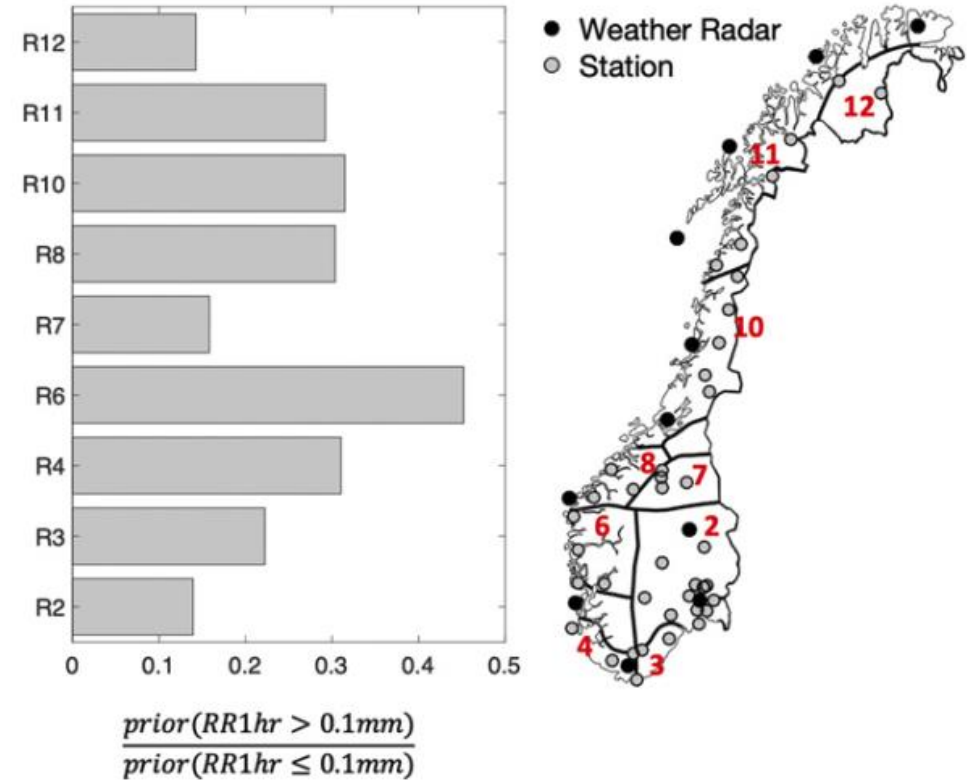


FIG. 1. (right) The locations of the 41 stations used in this study are depicted. They are grouped into 9 of the 13 precipitation regions. The locations of weather radars in Norway are also shown. (left) The ratio of the prior probability of precipitation to that of no precipitation in each region.

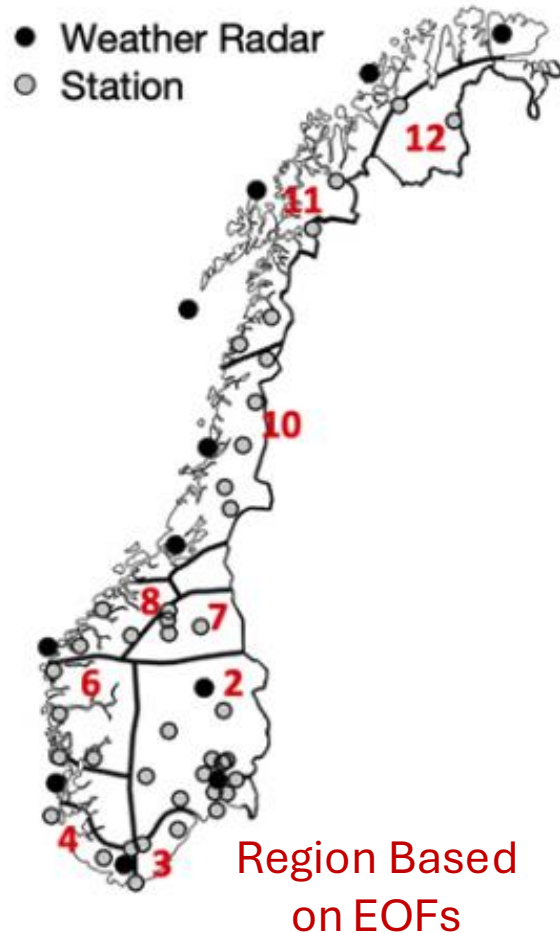
Predictors

TABLE 1. List of all variables considered as predictors for the RF model.

	Full name	Abbreviation
1	Accumulated precipitation in 1 h from the radar nowcasts	RR1hr(Radar)
2	Duration of precipitation in 1 h from the radar nowcasts	tL(Radar)
3	Accumulated precipitation in 6 h from AROME	RR6hr(NWP)
4	Accumulated precipitation in 1 h from AROME	RR1hr(NWP)
5	Air pressure at sea level	SLP
6	Air temperature at 2 m	T2m
7	Fog area fraction	Fog AF
8	Low-type cloud area fraction	LowC AF
9	Medium-type cloud area fraction	MediumC AF
10	High-type cloud area fraction	HighC AF
11	Relative humidity at 2 m	RH2m
12	Zonal wind U at 10 m	U10m
13	Meridional wind V at 10 m	V10m
14	Atmospheric boundary layer thickness	ABLT
15	Wind speed normal to the topographic aspect	WSNT
16	Average dewpoint depression from 1000 to 500 mb	DD(1000–500 mb)
17	K index	K

2.5km res. NWP
covering Scandinavia
and the Nordic Sea

Predictors



Local Predictions

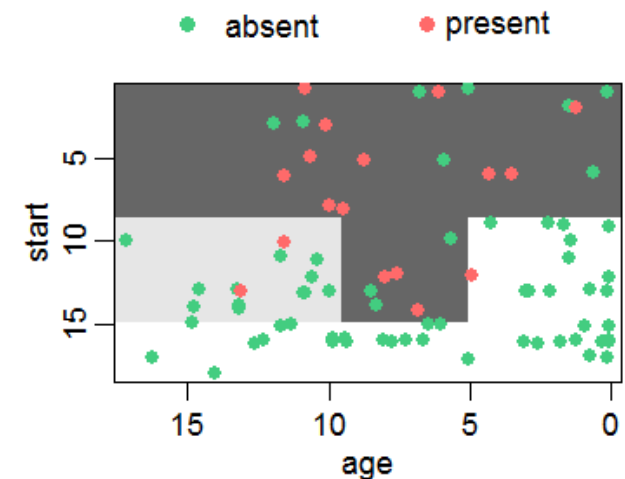
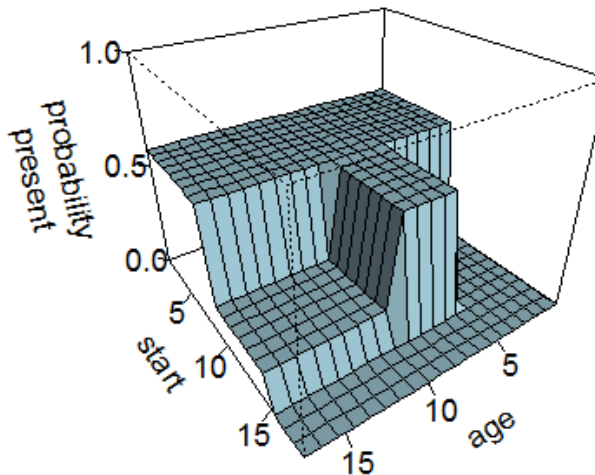
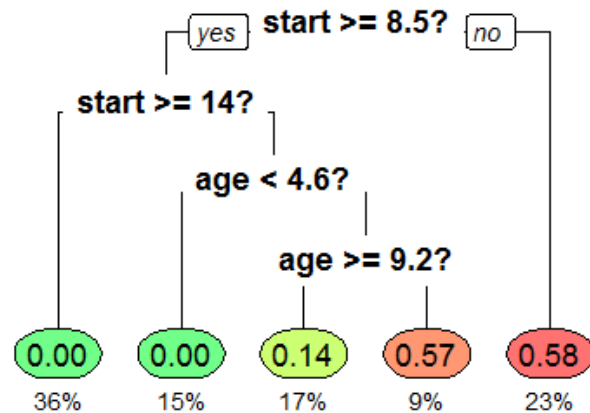
$$\begin{cases} p_1 = p_c, & \text{if } d(x_c, x_i) = 0, \\ p_2 = \frac{\sum_{i=1}^N w_i p_i}{\sum_{i=1}^N w_i}, & \text{if } d(x_c, x_i) > 0, \end{cases} \quad (1)$$

Inv. distance weighted of surroundings (100km x 100km)

Predictors are the average of p1 and p2

Methods

- Random Forests (RF)
 - Collection of binary trees
 - Divide the dataset where it maximizes the “information gained”
 - Measured by entropy, **Gini index (used in this study)**, velc.(? “Or” version of “etc.”?)
 - Aggregate the predictions from many trees



(Example from wiki)

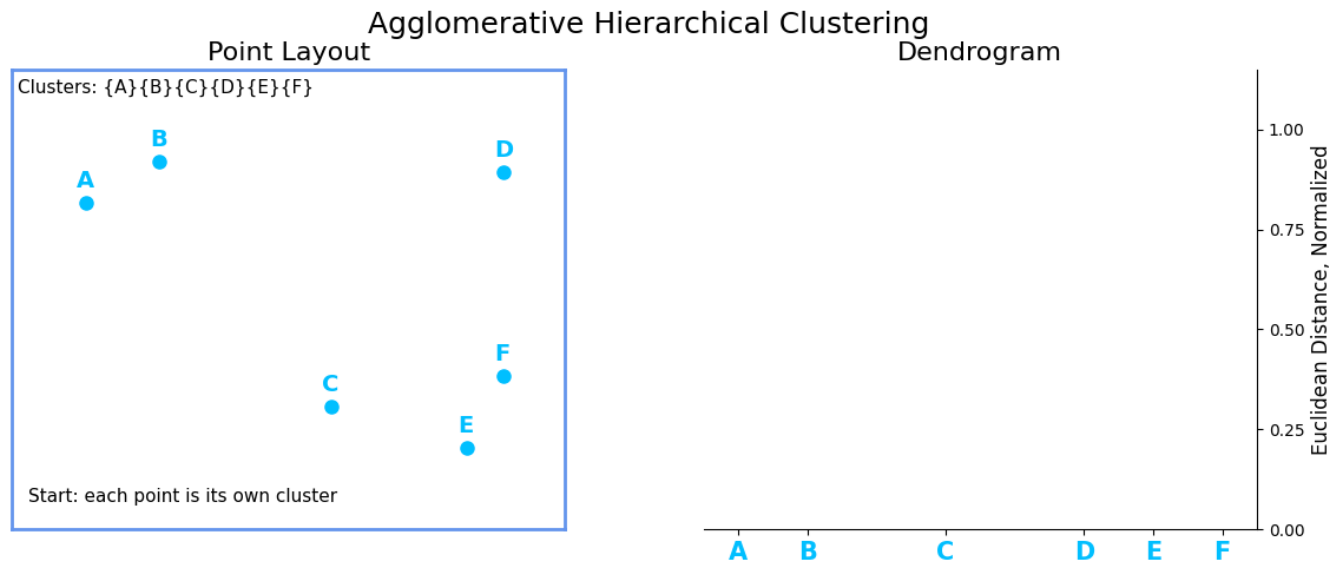
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-
- | | |
|--------------------------------------|------------------------------------|
| + Naturally tolerates missing values | - Bias toward the dominant feature |
| + Robust to outliers | - Poor extrapolation |
| + No preprocessing needed | |

The paper also presents another method for determining feature importance, which I skipped (too many figures)

Methods

- Hierarchical Clustering
 - Feature importance
 - Group predictors by hierarchical clustering of spearman corr. matrix
 - Shuffling the values of one or more predictors and observing their predictions drop



This paper will use
Spearman corr. coeff.
as a distance measure
(I assume they are
using $\text{abs}(\rho)$)

(Example from wiki)

Evaluation Metrics

TABLE 3. Descriptions of various metrics used in this study.

Abbreviation	Full name	Meanings	Best	Worst
POD	Probability of detection	The fractional success of predicting precipitation out of all occasions when precipitation is observed	1	0
POFD	Probability of false detection	The number of times precipitation is falsely predicted out of all occasions when precipitation is not observed	0	1
FAR	False alarm ration	The number of times precipitation is falsely predicted out of all precipitation predictions	0	1
SR	Success ratio	The fractional success of precipitation prediction out of all precipitation predictions	1	0
CSI	Critical success index	The fractional success of precipitation prediction out of all precipitation predictions and missed predictions of precipitation	1	0
SS0	Skill score relative to a base prediction	The relative improvement of the classification model from the base prediction of always predicting the majority Class 0 (no precipitation)	1	<0
bias	Frequency bias	Quantifies whether the RF model tends to predict the precipitation more (bias > 1) or less (bias < 1) often than it is actually observed	1	

Actually, this is a good summary of binary evaluation metrics

Evaluation Metrics (In case words are too hard)

TABLE 2. A 2×2 contingency table for a binary classification; A , B , C , and D represent the number of cases belonging to each category.

	Observe 1	Observe 0
Predict 1	A (hit)	B (false alarm)
Predict 0	C (miss)	D (correct rejection)

$$\text{ACC} = \frac{A + D}{A + B + C + D},$$

$$\text{FAR} = \frac{B}{A + B},$$

$$\text{SS0} = \frac{\text{ACC} - \text{ACC}(0)}{1 - \text{ACC}(0)},$$

$$\text{POD} = \frac{A}{A + C},$$

$$\text{SR} = \frac{A}{A + B},$$

$$\text{bias} = \frac{A + B}{A + C},$$

$$\text{POFD} = \frac{B}{B + D},$$

$$\text{CSI} = \frac{A}{A + B + C},$$

Also, check **Roebber 2009** for more info

Evaluation Metrics

NOTES AND CORRESPONDENCE

Visualizing Multiple Measures of Forecast Quality

PAUL J. ROEBBER

Atmospheric Science Group, Department of Mathematical Sciences, University of Wisconsin—Milwaukee, Milwaukee, Wisconsin

(Manuscript received 19 May 2008, in final form 24 July 2008)

ABSTRACT

A method for visually representing multiple measures of dichotomous (yes–no) forecast quality (probability of detection, false alarm ratio, bias, and critical success index) in a single diagram is presented. Illustration of the method is provided using performance statistics from two previously published forecast verification studies (snowfall density and convective initiation) and a verification of several new forecast datasets: Storm Prediction Center forecasts of severe storms (nontornadic and tornadic), Hydrometeorological Prediction Center forecasts of heavy precipitation (greater than 12.5 mm in a 6-h period), National Weather Service Forecast Office terminal aviation forecasts (ceiling and visibility), and medium-range ensemble forecasts of 500-hPa height anomalies. The use of such verification metrics in concert with more detailed investigations to advance forecasting is briefly discussed.

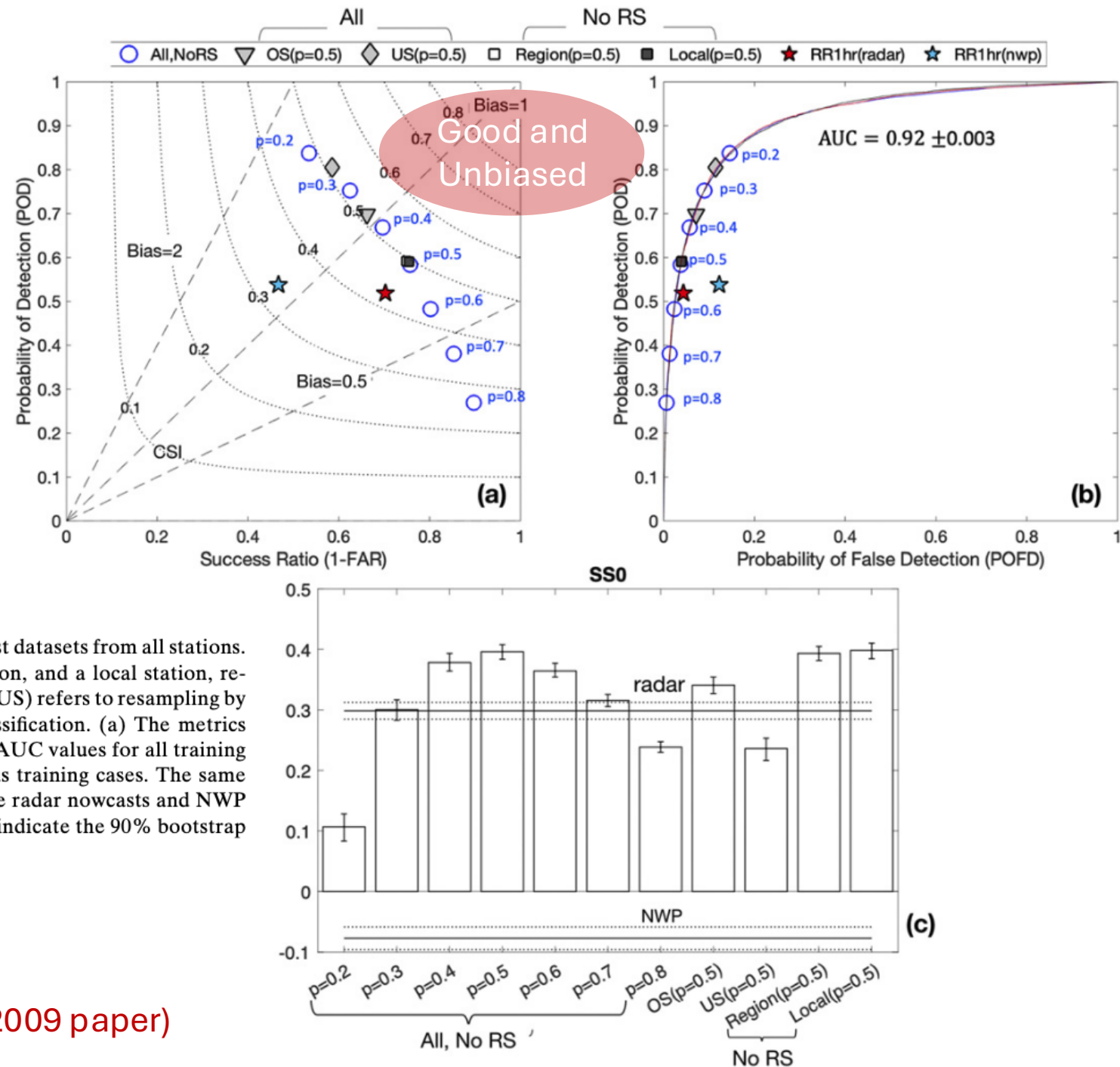
<https://doi.org/10.1175/2008WAF2222159.1>

Also, check **Roebber 2009** for more info



Discussion

Model Performance



(More info about this type of plot in Roebber 2009 paper)

Discussion

Model Performance

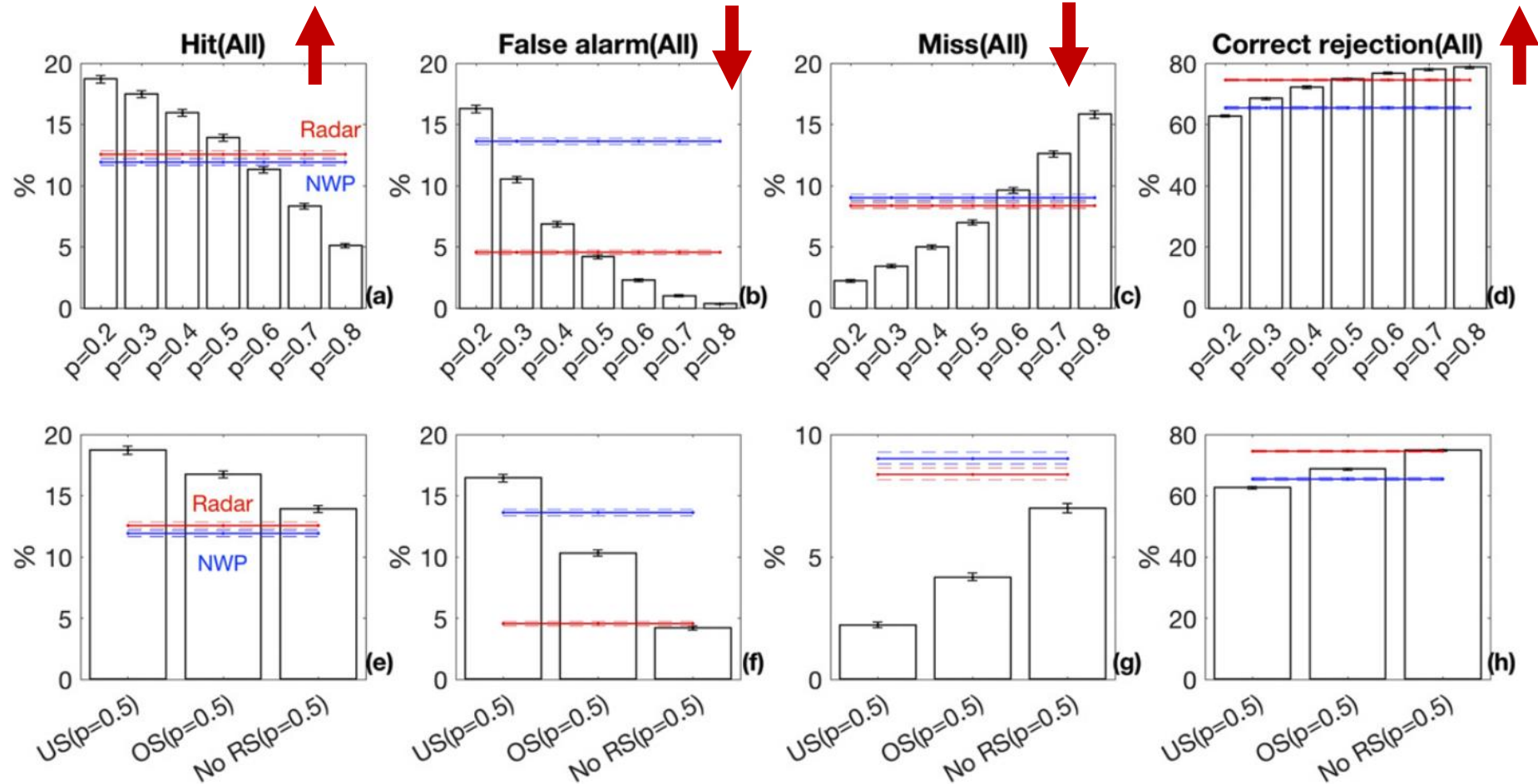
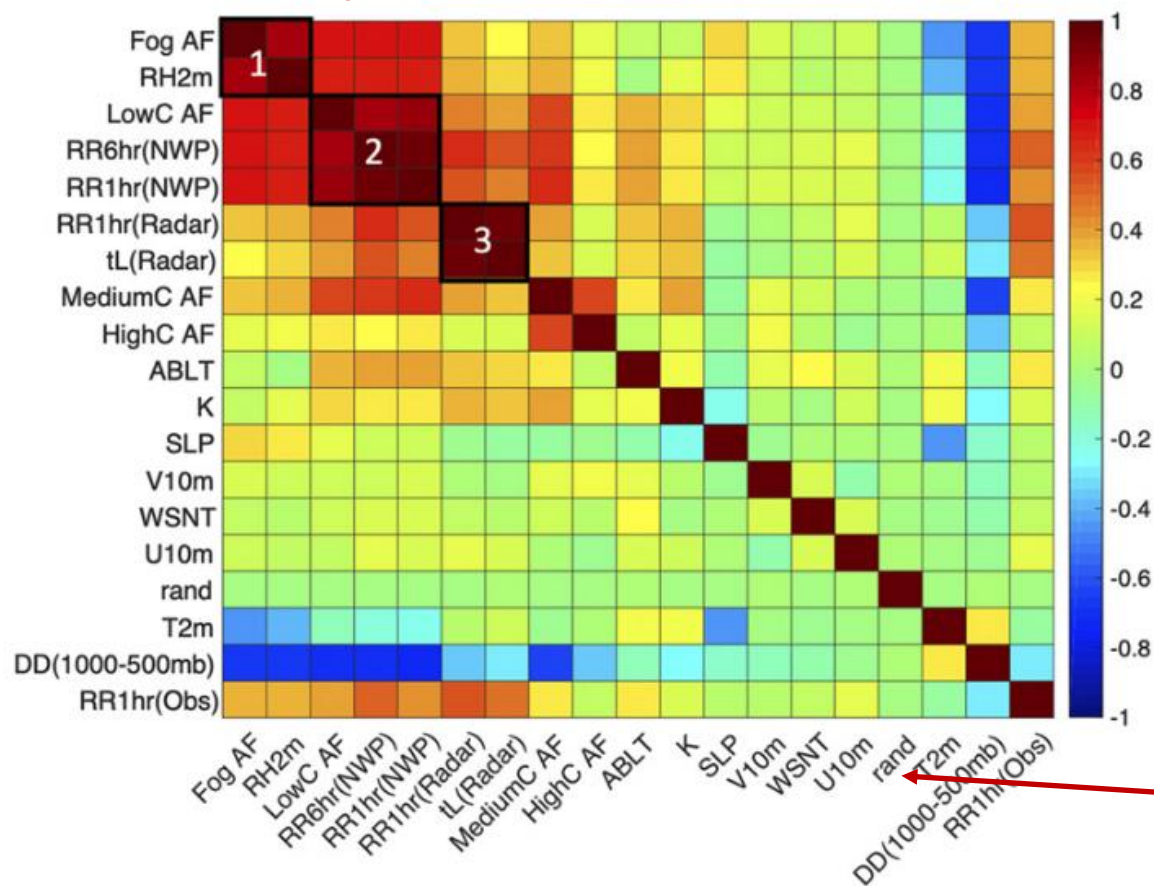


FIG. 4. (top) The percentage of test cases belonging to hit, false alarm, miss, and correct rejection for the RF model trained by using the original data (without resampling) from all stations but with different threshold probability p for classification. (bottom) The percentage of test cases belonging to hit, false alarm, miss, and correct rejection for the RF models trained by using data from all stations with and without resampling (No RS). OS and US refer to oversampling and undersampling, respectively. The red and blue solid lines indicate the values of the benchmarks: the radar nowcasts and NWP, respectively. The error bars and dashed lines indicate the corresponding 90% bootstrap confidence intervals for the RF models and benchmarks.

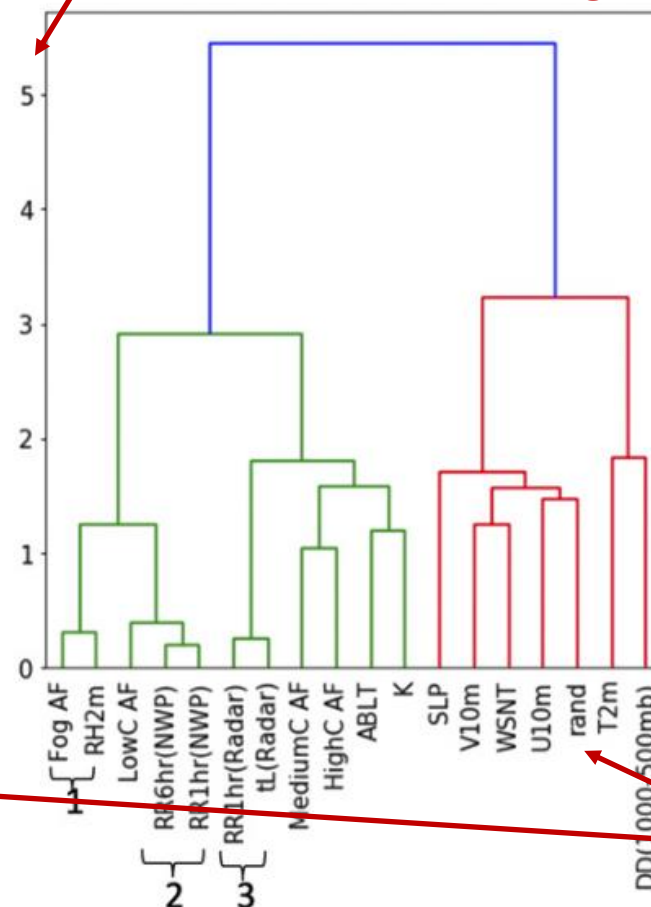
Discussion Feature Importance

Spearman Correlation Coefficient



I don't know why this is above 1... maybe some normalization is performed?

Hierarchical Clustering



Random value
Ideally, this should have the lowest importance

FIG. 7. (left) The heat map of all predictors listed in Table 1 and the precipitation observations for the dataset consisting of data from all stations. (right) The dendrogram used to visualize the hierarchical clustering of the Spearman correlations of the predictors. The three clusters of variables with strong collinearity are marked.

Discussion

Feature Importance

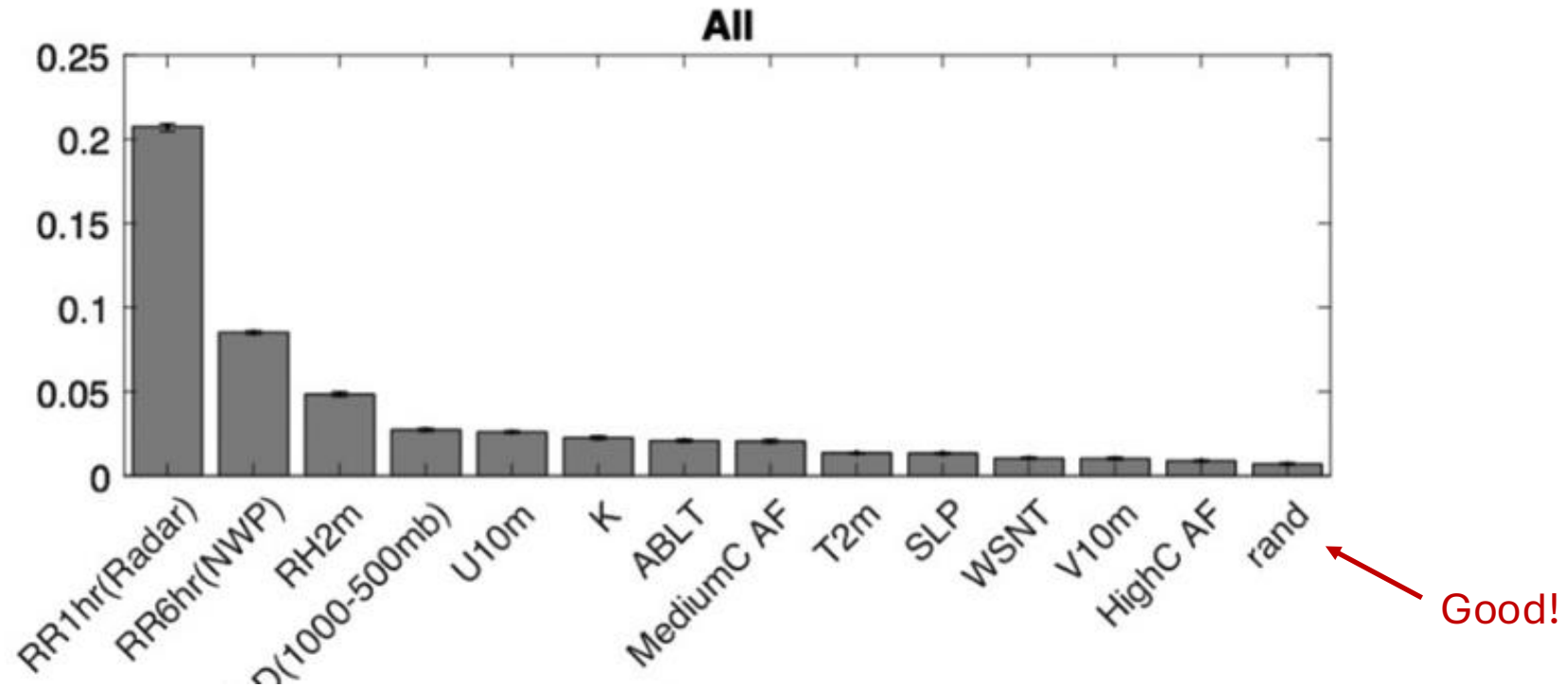


FIG. 8. Ranking importance by permutations of uncorrelated predictors for the RF models trained by using the training datasets from (top) all stations and (others) from stations in each region. The error bars indicate the range between the minimum and maximum values of importance ranking obtained by repeating the process 10 times.

Discussion

Feature Importance

